

Project: Remote Work Survey Analysis (2020)

SQL ANALYSIS PHASE SPECIFICATION

Objective

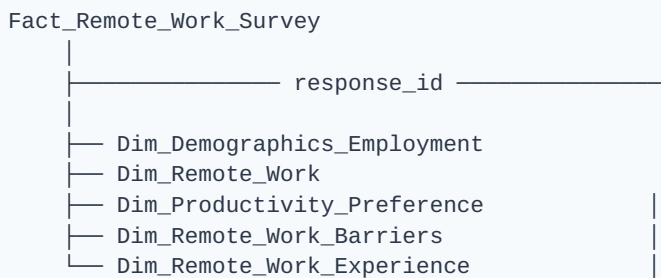
The SQL analysis phase aims to answer real-world business questions using the cleaned survey data stored in PostgreSQL. Unlike Exploratory Data Analysis (EDA), which focuses on understanding individual datasets, SQL analysis combines multiple tables to generate meaningful business insights through filtering, aggregation, grouping, and joins. The results of these SQL queries will be used as the foundation for the Power BI dashboards.

SQL Analysis Workflow

1. Import cleaned datasets into PostgreSQL.
2. Create relationships between the fact table and dimension tables using `response_id`.
3. Write business-focused SQL queries.
4. Generate actionable insights from the query results.
5. Connect PostgreSQL to Power BI for interactive dashboard creation.

Database Schema

The project follows a star-schema design centered around the survey responses:



Note: Each dimension table is connected to the fact table using the `response_id` field.

Goals of SQL Analysis

The SQL analysis is designed to answer core business questions including:

- Who participated in the survey?
- How widely was remote work adopted?
- Which industries were most affected?
- How productive were employees while working remotely?
- What barriers did employees experience?
- Which remote work experiences were viewed positively or negatively?
- What relationships exist between demographics, productivity, and remote work preferences?

Business Analysis Sections

1. Demographics & Workforce Analysis

Purpose: Understand the characteristics of survey participants.

- How many employees participated in the survey?
- What is the gender and age distribution?
- Which industries have the highest participation?
- What are the most common occupations?
- What company sizes are represented?
- How many respondents are managers vs. individual contributors?
- What is the distribution of job tenure & household types?
- Which regions have the highest participation?

2. Remote Work Adoption Analysis

Purpose: Understand how organizations implemented remote work before and during COVID-19.

- How common was remote work before vs. during COVID-19?
- Did employers encourage remote work?
- Were organizations prepared for remote work?
- What was the most common remote work frequency?
- Which collaboration tools were most frequently used?
- Would employees recommend remote work?

3. Productivity & Work Preference Analysis

Purpose: Measure employee productivity and future workplace preferences.

- Which work model do employees prefer after COVID?
- Do employees feel more productive while working remotely?
- How much commuting time was saved?
- How do working hours compare between office and remote work?
- How much personal, family, and caring time changed?
- Does employer support influence future work preferences?

4. Remote Work Barriers Analysis

Purpose: Identify the major challenges employees faced while working remotely.

- What is the biggest remote work barrier?
- Which barriers are ranked second and third?
- Which barriers appear most frequently across all rankings?
- Which barriers are considered least significant?
- What challenges should organizations prioritize?

5. Remote Work Experience Analysis

Purpose: Understand what employees liked and disliked about remote work.

- What is the best and worst remote work experience?
- Which positive experiences appear most frequently?
- Which negative experiences occur most often?
- What should organizations continue after COVID?
- Which employee concerns require improvement?

6. Cross-Table Business Analysis

Purpose: Generate deeper insights by combining multiple dimension tables.

- Does gender influence remote work productivity?
- Which industries prefer hybrid work?
- Which company sizes provide the greatest employer support?
- Are managers more productive than non-managers?
- Does employer encouragement improve productivity?
- Which industries experience the most remote work barriers?
- Which age groups prefer full-time remote work?
- Does remote work frequency affect work-life balance?
- Do positive experiences increase future remote work preference?
- Which demographics experience the greatest challenges?

Expected SQL Concepts

Applied Concepts: SELECT, WHERE, ORDER BY, GROUP BY, HAVING | **Aggregations:** COUNT, AVG, MIN, MAX | **Logic & Joins:** CASE WHEN, INNER JOIN, LEFT JOIN | **Advanced:** Common Table Expressions (CTEs), Window Functions

Deliverables

- 30+ business-focused SQL queries
- Clear business insights for each query

- PostgreSQL database with a normalized schema
- Power BI connected directly to PostgreSQL
- Interactive dashboards built from SQL-driven data
- End-to-end analytics project demonstrating Excel, Python, PostgreSQL, SQL, and Power BI skills